

1. Executive Summary & Purpose

SafeDig is an automated, AI-augmented safety validation platform engineered to prevent utility strikes during civil engineering excavations in the United Kingdom. Under UK statutory frameworks (HSG47 and CDM 2015), striking buried infrastructure results in fatal injuries, gas explosions, electric flashovers, and major liabilities. SafeDig re-validates upstream utility warning disclosures against physical engineering drawings with ZERO ESCAPED HAZARDS.

NON-NEGOTIABLE DOMAIN INVARIANT (HSG47 COMPLIANCE)

SAFE_MODE is permanently active. Every detected hazard is escalated to a certified human engineer.

Every ambiguous, corrupted, or contradictory document fails toward HUMAN_REVIEW or BLOCKED, never toward AUTO_CLEAR.

The system never trades a false negative (escaped hazard) for a cleaner false-positive statistic.

2. The 10-Stage Pipeline Flow

Stage	Primary Component	Operational Function & Safety Scope
Stage 1: Ingestion	src/utils/paths.py	Self-healing path resolution, parses master inquiry index.xl...
Stage 2: Inventory	src/ingestion/inventory.py	Deterministic 1-to-1 matching. Separates technical maps from...
Stage 3: PDF Intake	src/domain/document.py	Inspects file magic, byte streams, and determines modality: ...
Stage 4: Knowledge	src/domain/legend.py	Resolves statutory provider (UKPN, SGN, Cadent), loads warni...
Stage 5: Spatial AOI	src/spatial/aoi_detector.py	Extracts excavation site polygon (Area of Interest), validat...
Stage 6: Dual CV Scan	src/vision/detector.py	PyMuPDF vector path geometric extraction + OpenCV 300 DPI ra...
Stage 7: Reconciliation	src/spatial/reconciler.py	Computes Shapely 2D spatial intersection between AOI buffer ...
Stage 8: 17 Gates	src/policy/gates.py	Evaluates all 17 mandatory release gates. Prevents auto-clea...
Stage 9: Advisory AI	src/orchestration/nodes.py	LangGraph executes Ollama LLM (Qwen 2.5) for operator summar...
Stage 10: Audit Persist	safedig.db (SQLAlchemy)	Generates high-res visual evidence package and locks SHA-256...

3. Spatial Reconciliation & Buffer Mechanics

SafeDig does not rely on simple bounding boxes. It implements millimeter-accurate 2D polygon math via Shapely:

- 1. Area of Interest (AOI): Extracted from site redline boundaries, verified closed (coords[0] == coords[-1]).
- 2. Statutory Safety Buffers: SafeDig applies a 5.0m buffer for standard low-voltage and distribution pipelines, and a statutory 15.0m buffer for Cadent / SGN High-Pressure gas transmission pipelines (HSG47).
- 3. Geometric Intersection: Computes polygon.intersects(vector_line) and polygon.distance(vector_line).
- 4. Reconciliation Matrix: Classifies interaction between upstream enquiry disclosures and physical drawing findings.

CONFIRMED_CLEAN	Upstream: 'No Assets' CV: Zero Intersecting Vectors	AUTO_CLEAR
Both sources confirm complete absence of underground assets inside excavation buffer. Clearance approved.		
MATCH (Low/Med)	Upstream: 'Assets Present' CV: Vectors Intersect AOI	HUMAN_REVIEW
Detected underground utility confirmed. Surfaced to engineer for hand-dig permit condition assignment.		
MATCH (High/Crit)	Upstream: 'Assets Present' CV: 11kV/33kV HV or HP Gas	HUMAN_REVIEW
CRITICAL LIFE-SAFETY: Gate 12 immediately blocks auto-clearance. Requires formal engineer sign-off.		
MISSED_WARNING	Upstream: 'No Assets' CV: Live Utility Vectors Inside AOI	HUMAN_REVIEW
PRIMARY RESCUE: Catches dangerous omissions where utility letters claimed clear but drawings contain live mains.		

4. Dual-Engine Spatial Asset Extraction

SafeDig solves cartographic complexity by fusing two independent detection channels:

- Channel A (PyMuPDF Vector Parsing): Extracts native PDF path objects directly from binary streams. Yields exact coordinate geometries, stroke widths, and RGB values with zero pixelization or blur.
- Channel B (OpenCV 300 DPI Computer Vision): Renders high-resolution bitmaps to analyze legacy scanned plans, raster textures, and rasterized hatching patterns that lack native PDF vector paths.

The reconciliation engine fuses both inventories into a single verified geospatial coordinate frame.

5. Interactive QA Workspace & Evidence Dossier

- When an automated clearance is held for human review, the job is dispatched to the SafeDig QA Workspace UI:
- High-Resolution Map Inspection: Interactive viewport featuring hardware-accelerated canvas pan & zoom (50%-600%).
 - Animated Radar Loader: Real-time visual feedback indicating map image and evidence bundle loading.
 - 17 Gates Live Status Panel: Visual badge breakdown showing exact pass/fail reasons for each gate.
 - Forensic Evidence Package: Pre-crops the excavation zone (AOI crop), full sheet context, and GeoJSON vectors.
 - Human Override Audit Trail: Engineers can confirm findings or record mitigation steps with cryptographic timestamps.

6. Enterprise Production Deployment Architecture

Containerization	Docker & Multi-Stage Compose
Hardened Debian-slim containers bundling PyMuPDF, OpenCV, and FastAPI.	
Reverse Proxy	Nginx Reverse Proxy
SSL termination, rate limiting, static asset caching, and request buffering.	
Process Manager	Systemd / Windows Service
safedig.service background daemon with automatic restart on failure.	
Data Persistence	SQLAlchemy + SQLite / Postgres
safedig.db for local edge deployments; PostgreSQL for enterprise clustering.	
Cross-Machine Ready	Self-Healing Path Resolvers
Dynamic directory translation ensures jobs run seamlessly across Windows/Linux.	